

Chapter One

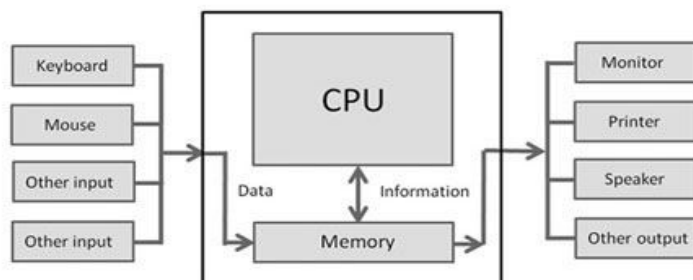
1.1 Introduction to computer

Definition of computer

What is Computer?

The term computer is derived from the Latin term 'computare', this means to calculate or programmable machine. **Computer** is an electronic device that is designed to work with Information. **IT cannot do anything without a Program**. It represents the decimal numbers through a string of binary digits. The Word 'Computer' usually refers to the **Central Processor Unit** plus **Internal memory**.

The computer is an electronic device that takes input from the user and processes these data under the control of a set of instructions (called program) and gives the result (output) and saves future use. It can process both numerical and non-numerical (arithmetic and logical) calculations. A **computer** is "a programmable electronic device designed to accept data, perform prescribed mathematical and logical operations at high speed, and display the results of these operations."



The basic components of a modern digital computer are: **Input Device**, **Output Device**, **Central Processor** and **memory**. Four Functions about computer are:

Accepts data	Input
Processes data	Processing
Produces output	Output
Stores results	Storage

Input (Data): Input is the raw information entered into a computer from the **input devices**. It is the collection of letters, numbers, images etc.

Process: Process is the operation of data as per given instruction. It is totally internal process of the computer system.

Output: Output is the processed data given by computer after data processing. Output is also called as Result. We can save these results in the storage devices for the future use.

History of Computers

The first counting device was used by the primitive people. They used sticks, stones and bones as counting tools. As human mind and technology improved with time more computing devices were developed. Some of the popular computing devices starting with the first to recent ones are described below;

Abacus

The history of computer begins with the birth of abacus which is believed to be the first computer. It is said that Chinese invented Abacus around 4,000 years ago.

It was a wooden rack which has metal rods with beads mounted on them. The beads were moved by the abacus operator according to some rules to perform arithmetic calculations. Abacus is still used in some countries like China, Russia and Japan. An image of this tool is shown below;



Napier's Bones

It was a manually-operated calculating device which was invented by John Napier (1550-1617) of Merchiston. In this calculating tool, he used 9 different ivory strips or bones marked with numbers to multiply and divide. So, the tool became known as "Napier's Bones. It was also the first machine to use the decimal point.

Pascaline

Pascaline is also known as Arithmetic Machine or Adding Machine. It was invented between 1642 and 1644 by a French mathematician-philosopher Blaise Pascal. It is believed that it was the first mechanical and automatic calculator.

Pascal invented this machine to help his father, a tax accountant. It could only perform addition and subtraction. It was a wooden box with a series of gears and wheels. When a wheel is rotated one revolution, it rotates the neighboring wheel. A series of windows is given on the top of the wheels to read the totals. An image of this tool is shown below;



ENIAC

- ♣ ENIAC stands for Electronic Numerical Integrator and Computer.
- ♣ It was the first electronic general purpose computer.
- ♣ Completed in 1946.
- ♣ Developed by John Presper Eckert and John W. Mauchly.

UNIVAC 1

- ♣ The UNIVAC I (UNIVersal Automatic Computer 1) was the first commercial computer.
- ♣ Designed by J. Presper Eckert and John Mauchly.

EDVAC

- ♣ EDVAC stands for Electronic Discrete Variable Automatic Computer
- ♣ The First Stored Program Computer • Designed by Von Neumann in 1952.
- ♣ It has a memory to hold both a stored program as well as data.

Generations of computer

Generations of Computer

A generation of computers refers to the specific improvements in computer technology with time. In 1946, electronic pathways called circuits were developed to perform the counting. It replaced the gears and other mechanical parts used for counting in previous computing machines.

In each new generation, the circuits became smaller and more advanced than the previous generation circuits. The miniaturization helped increase the speed, memory and power of computers. There are five generations of computers which are described below;

First Generation Computers

The first generation (1940-1956) computers were slow, huge and expensive. In these computers, vacuum tubes were used as the basic components of CPU and memory. These computers were mainly depended on batch operating system and punch cards. Magnetic tape and paper tape were used as output and input devices in this generation;

Some of the popular first generation computers are;

- ♣ **ENIAC** (Electronic Numerical Integrator and Computer)
- ♣ **EDVAC** (Electronic Discrete Variable Automatic Computer)
- ♣ **UNIVACI**(Universal Automatic Computer)
- ♣ **IBM-701**
- ♣ **IBM-650.**

Second Generation Computers

The second generation (1959-1965) was the era of the transistor computers. These computers used transistors which were cheap, compact and consuming less power; it made transistor computers faster than the first generation computers.

In this generation, magnetic cores were used as the primary memory and magnetic disc and tapes were used as the secondary storage. Assembly language and programming languages like COBOL and FORTRAN, and Batch processing and multiprogramming operating systems were used in these computers.

Some of the popular second generation computers are;

- ♣ **IBM 1620**
- ♣ **IBM 7094**
- ♣ **CDC 1604**
- ♣ **CDC 3600**
- ♣ **UNIVAC 1108**

Third Generation Computers

The third generation computers used integrated circuits (ICs) instead of transistors. A single IC can pack huge number of transistors which increased the power of a computer and reduced the cost. The computers also became more reliable, efficient and smaller in size. These generation computers used remote processing, time-sharing, multi programming as operating system. Also, the high-level programming languages like FORTRON-II TO IV, COBOL, PASCAL PL/1, ALGOL-68 were used in this generation.

Some of the popular third generation computers are;

- ♣ **IBM-360 series**
- ♣ **Honeywell-6000 series**
- ♣ **PDP(Personal Data Processor)**
- ♣ **IBM-370/168**
- ♣ **TDC-316**

Fourth Generation Computers

The fourth generation (1971-1980) computers used very large scale integrated (VLSI) circuits; a chip containing millions of transistors and other circuit elements. These chips made this generation computers more compact, powerful, fast and affordable. These generation computers used real time, time sharing and distributed operating system. The programming languages like C, C++, DBASE were also used in this generation.

Some of the popular fourth generation computers are;

- ♣ **DEC 10**
- ♣ **STAR 1000**
- ♣ **PDP 11**
- ♣ **CRAY-1(Super Computer)**
- ♣ **CRAY-X-MP(Super Computer)**

Fifth Generation Computers

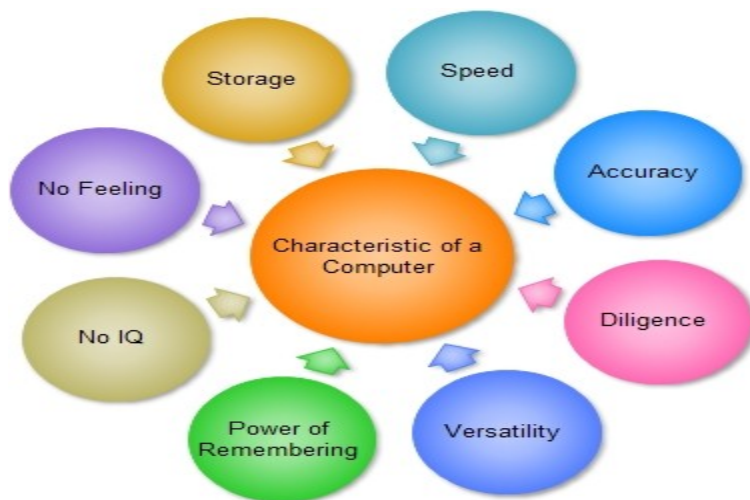
In fifth generation (1980-till date) computers, the VLSI technology was replaced with ULSI (Ultra Large Scale Integration). It made possible the production of microprocessor chips with ten million electronic components. This generation computers used parallel processing hardware and AI (Artificial Intelligence) software. The programming languages used in this generation were C, C++, Java, .Net, etc.

Some of the popular fifth generation computers are;

- ♣ Desktop
- ♣ Laptop
- ♣ NoteBook

1.2 Characteristic of a Computer

Basic characteristics about computer are



Speed: – As you know computer can work very fast. It takes only few seconds for calculations that we take hours to complete. You will be surprised to know that computer can perform millions (1,000,000) of instructions and even more per second.

Therefore, we determine the speed of computer in terms of microsecond (10⁻⁶ part of a second) or nanosecond (10 to the power -9 part of a second). From this you can imagine how fast your computer performs work.

Accuracy: – The degree of accuracy of computer is very high and every calculation is performed with the same accuracy. The errors in computer are due to human and inaccurate data.

Diligence: – A computer is free from tiredness, lack of concentration, fatigue, etc. It can work for hours without creating any error. If millions of calculations are to be performed, a computer will perform every calculation with the same accuracy. Due to this capability it overpowers human being in routine type of work.

Versatility: – It means the capacity to perform completely different type of work. You may use your computer to prepare payroll slips. Next moment you may use it for inventory management or to prepare electric bills.

5. Power of Remembering: – Computer has the power of storing any amount of information or data. Any information can be stored and recalled as long as you require it, for any numbers of years. It depends entirely upon you how much data you want to store in a computer and when to lose or retrieve these data.

6. No IQ: – Computer is a dumb machine and it cannot do any work without instruction from the user. It performs the instructions at tremendous speed and with accuracy. It is you to decide what you want to do and in what sequence. So a computer cannot take its own decision as you can.

7. No Feeling: – It does not have feelings or emotion, taste, knowledge and experience. Thus it does not get tired even after long hours of work. It does not distinguish between users.

8. Storage: – The Computer has an in-built memory where it can store a large amount of data. You can also store data in secondary storage devices such as floppies, which can be kept outside your computer and can be carried to other computers.

1.3 Types of Computer

Classification of Computers on the Basis of Size and Capacity

1. SuperComputer
2. Mainframe Computer
3. Mini Computer
4. Micro Computer

Computers on the Basis Purpose

1. General Purpose

2. Special Purpose

Computers based on Hardware Design and Data Handling.

1. Analog Computer

2. Digital Computer

3. Hybrid Computer

Classifications of Computers based on Hardware Design and Data Handling.

A. Analogue computer

- ✚ **An Analog computer** is performing duties to using continuous data. These computers represent variables by physical quantities.
- ✚ Analog computers are fundamentally used to gauge actual units which solved voltage, pressure, electric flow, temperature, and convert them into digits.
- ✚ These computers are very fast in their operations to be carried and a powerful tool to solve differential equations.
- ✚ The productivity of this computer increases, when we get the consequence of the information as diagrams analog Computers can't store measurements.

B. Digital computer

- ✚ A Digital computer represents physical quantities with the help of digits and numbers.
- ✚ These numbers are used to perform Mathematics calculations and made logical decision to reach a conclusion, depending on, the data they receive from the user.
- ✚ The digital computer is also known as the Digital Data processing system. It's performs such as addition, occurrence, multiplication, or calculations as well as logical operations.
- ✚ It can also perform organize and analyze data, control industrial and other processes such as global weather patterns.
- ✚ These types of computer that stores and processes data in the digital form, usually in the binary 0 or 1 system.

C. Hybrid computer

- ✚ A hybrid computer is a combination of analog and digital computer which built for using both data by a single control system.
- ✚ It encompasses the best features of both Analog and Digital computers.

- ✚ This computer is capable of inputting those signals and converts them to digital form.
- ✚ The purpose of designing this computer is to provide functions and features that could found on both devices. It made to solve extremely complex calculations or problems.
- ✚ This computer system setup offers a cost-effective method of performing complex simulations.
- ✚ These computers are mostly used in very large industries, organizations, and business firms.

Computers on the Basis of Size and Capacity

1. SuperComputer
2. Mainframe Computer
3. Mini Computer
4. Micro Computer

1. Micro computer

- ✚ The design of microcomputers is minimal in size and storage capacity.
- ✚ Today we are using many computers at home is also the most common microcomputer.
- ✚ The primary purpose of microcomputers is to keep and process the everyday tasks and needs of the people.
- ✚ Only one person can work on a single PC at a time, but its operating system is **multitasking**. The PC can be connected to the Internet to take benefits and enhance the user experience.
- ✚ Examples: **Desktops, tablets, smartphones, and Laptops, workstation, server, and palmtop**

2. Mini Computer

- ✚ Minicomputer is a **digital and multi-user** computer system with the connection of more than one CPU. Thus, many people can work on these computers simultaneously instead of a single person. Also, it can process with other accessories like a printer, plotter, etc.
- ✚ Minicomputers are the medium type of computers that have more functionality power and are expensive than microcomputers. On the other hand, the size, storage, and speed of minicomputers are **large but less than** the mainframe and supercomputers.

- ✚ Minicomputers are made for performing multiple computing tasks at a single point of time, instead of assigning many microcomputers for a single task, which will be time-consuming and expensive.
- ✚ In general, a minicomputer is a multi-threaded system (several processes at a time) capable of supporting from one to up to 200 users simultaneously: these computers are currently used to *store large databases, multi-user applications, and the automation industry*.
- ✚ Example **PDP 11** and **IBM (8000 Series)**.

3. Mainframe Computer

- ✚ Mainframe computers are **multi-programming, high-performance** computers, and multi-user, which mean they can handle the workload of more than 100 users at a time on the computer.
- ✚ The storage capacity of the mainframe is enormous, with a high-speed data process as well. As well as handling hundreds of input and output devices at a time.
- ✚ The mainframe is a highly efficient computer capable of **simultaneously solving complex calculations and continuously for a long time**. These computers have several microprocessors that have the ability to function the data at too high performance and speed.
- ✚ Mainframe computers are mainly used by departmental and commercial organizations like Banks, Companies, Scientific research centers, and governmental departments like railways. These computers can work for 24 hours. Hundreds of users can work on these computations simultaneously.
- ✚ Using the mainframe completes the tasks, Such as keeping details of payments, research centres, advertising, sending bills and notices, paying employees, ticket booking, maintaining details of purchases by users, keeping detailed tax details, etc.
- ✚ **Examples**: IBM Es000 series, ICL39 Series, and CDC 6600.

4. Super Computer

- ✚ A Supercomputer is the very **fastest** and **powerful**, and **expensive** type of computer for processing data. Supercomputers' size and storage capacity are also huge (can occupy huge premises) designed to process vast amounts of data in a short time with high productivity.
- ✚ These are specially made to perform **multi-specific tasks**. Therefore, many CPUs work in parallel order on these supercomputers. This function of a Supercomputer is called Multiprocessing or Parallel Processing

✚ **Examples:** IBM Roadburner, *IBM Blue Gene*, *Intel ASI Red*, PARAM-1000,
and *CRAY-XMP-14*

Computers based on Purpose

1. General Purpose

- ✚ **General computers** can do various **everyday tasks** such as writing a word processing letter, Document preparation, recording, financial analysis, Printing documents, creating databases, and calculations with accuracy and consistency.
- ✚ The size, storage capacity, and cost of such computers are mainly less.
- ✚ The ability of these computers is limited in performing specialized tasks. Still, it has **versatility and useful** for serving people basic needs at home or in the workplace in the environment.

Examples: Desktops, laptops, smartphones, and tablets are used on daily basis for general purposes.

2. Special Purpose

- ✚ These computers are designed to perform a particular or specialized task.
- ✚ The size, storage capacity, and cost of such computers mainly depend on the nature and size of the work.
- ✚ The function of these computers is consistent with any particular task.
- ✚ The special computer **needs specific and input and devices** as well as a compatible mother board with the processor to conduct work efficiently.
- ✚ These computers are used for special purposes :.
 - ✓ *Weather forecasting, space research, agriculture, engineering, meteorology, satellite operation, traffic control, and research in chemical sciences.*

Examples:

- Automatic teller machines (ATM),
- Washing machines,
- Surveillance equipment,
- Weather-forecasting simulators,
- Traffic-control computers,
- Defense-oriented applications,
- Oil-exploration systems,

- Military planes controlling computers.

1.4 Applications of Computers

1.5.1 Generic applications

Computers play a role in every field of life. They are used in homes, business, educational institutions, research organizations, medical field, government offices, entertainment, etc.

Home

Computers are used at homes for several purposes like

- ✓ Online bill payment,
- ✓ Watching movies or shows at home,
- ✓ Home tutoring,
- ✓ Social media access,
- ✓ Playing games,
- ✓ Internet access
- ✓ They provide communication through electronic mail.
- ✓ They help to avail work from home facility for corporate employees.

Medical Field

Computers are used in hospitals

- ✓ to maintain a database of patients' history,
- ✓ diagnosis, X-rays,
- ✓ Live monitoring of patients, etc.
- ✓ Surgeons nowadays use robotic surgical devices to perform delicate operations, and conduct surgeries remotely.
- ✓ Virtual reality technologies are also used for training purposes.
- ✓ It also helps to monitor the fetus inside the mother's womb.

Entertainment

Computers help in entertainment in different ways:

- ✓ To watch movies online,
- ✓ play games online;
- ✓ act as a virtual entertainer in playing games,
- ✓ Listening to music, etc.

- ✓ MIDI instruments greatly help people in the entertainment industry in recording music with artificial instruments.
- ✓ Videos can be fed from computers to full screen televisions.
- ✓ Photo editors are available with fabulous features.

Industry

Computers are used to perform several tasks in industries

- ✓ Like managing inventory,
- ✓ designing purpose,
- ✓ creating virtual sample products,
- ✓ interior designing,
- ✓ Video conferencing, etc.

Education

Computers are used in education sector through

- ✓ online classes,
- ✓ online examinations,
- ✓ They help in increased use of audio-visual aids in the education field.
- ✓ Computers help the student community to avail online educational support.
- ✓ referring e-books,
- ✓ Online tutoring, etc.

Government

In government sectors, computers are used in

- ✓ data processing,
- ✓ maintaining a database of citizens and
- ✓ Supporting a paperless environment.
- ✓ The country's defence organizations have greatly benefitted from computers in their use for **missile development, satellites, rocket launches**, etc.

Banking

In the banking sector, computers are used

- ✓ To store details of customers and conduct transactions, such as withdrawal and deposit of money through ATMs.

- ✓ Banks have reduced manual errors and expenses to a great extent through extensive use of computers.

Business

Nowadays, computers are totally integrated into business.

- ✓ The main objective of business is transaction processing, which involves transactions with suppliers, employees or customers.
- ✓ Computers can make these transactions easy and accurate.
- ✓ People can analyse investments, sales, expenses, markets and other aspects of business using computers.

Training

- ✓ Many organizations use computer-based training to train their employees, to save money and improve performance.
- ✓ Video conferencing through computers allows saving of time and travelling costs by being able to connect people in various locations.

Arts

- ✓ Computers are extensively used in dance, photography, arts and culture.
- ✓ The fluid movement of dance can be shown live via animation.
- ✓ Photos can be digitized using computers.

Science and Engineering

- ✓ Computers with high performance are used to stimulate dynamic process in Science and Engineering.
- ✓ Supercomputers have numerous applications in area of Research and Development (R&D).
- ✓ Topographic images can be created through computers.
- ✓ Scientists use computers to plot and analyse data to have a better understanding of earthquakes

Environmental informatics

Environmental informatics is a field of applied computer science that develops and uses the techniques of information processing for environmental protection, research and engineering.

Environmental informatics (EI) is a developing field of science that applies information processing, management, and sharing strategies to the interdisciplinary field of environmental science. Applications include the integration of information and knowledge, the application of computational intelligence to environmental data, and the identification of the environmental impacts of information technology. EI helps scientists define information processing requirements, analyse real-world problems, and solve those problems using informatics methodologies and tools.

Measures of Computer Quality and Performance

computer performance is the amount of useful work accomplished by a computer system.

The two techniques to assess performance are:

1. **Computer speed** is one of the leading parameters in assessing a computer's performance. It can be compared to the horsepower of an engine.
 - The speed of a computer is measured by how quickly it completes a single task.
2. **The throughput metric** is used to determine how many jobs a computer can execute in a given amount of time.

Computer performance metrics (things to measure) include availability, response time, channel capacity, latency, completion time, service time, bandwidth, throughput, relative efficiency, scalability, performance per watt, compression ratio, instruction path length and speed up.

Advantage of Computers

Multitasking: A person can do multiple tasks, and multiple operations at the same time, and calculate numerical problems within a few seconds. The computer can perform millions or trillions of work in one second.

Speed: Now a day's computer has a vital role in human life. One of the most advantages of computers is their incredible speed, which helps human to finish their task in a few seconds.

Cost/ Stores huge: Amount of knowledge it's a coffee cost solution. A person can save huge data within a coffee budget. A centralized database for storing information is a major advantage that will reduce costs.

Accuracy: One of the essential advantages of computers is they will perform not only calculations but also with accuracy.

Data Security: Protecting digital data is understood as data security.

Task completer: Completes tasks that might be impossible for humans to complete.

Communication: The computer helps the user better understand and communicate with other devices.

Productivity: The level of productivity gets automatically doubled as the computer can do the work very fast.

Reduces workload: Information is often accessed by more than one person with the necessity for work to be duplicated.

Reliability: Computers can perform the same sort of work repeatedly without throwing up errors thanks to tiredness or boredom, which are quite common among humans.

Storage: The pc has an in-built memory where it can store an outsized amount of knowledge. You can also store data in auxiliary storage devices.

Disadvantages of computer

- **Virus and hacking attacks:** A virus may be a worm and hacking is just unauthorized access over a computer for a few illicit purposes. Viruses can go to another system from email attachments, viewing an infected website advertisement, through removable devices like USBs, etc.
- **Online Cyber Crimes:** Online [cyber-crime](#) means computers and networks may have been utilized in order to commit a crime.
- **Reduction in employed opportunity:** Mainly past generation hasn't used the pc or they need the knowledge of computers they faced an enormous problem when computers came into the field.
- **High Cost:** Computers are expensive. Even the foremost affordable computers are still very expensive for the typical person in South Africa. Since computers empower people.
- **Health Problems:** Prolonged use of computers can lead to various health Hazards. Too much sitting near the screen results in eye strain and drying up of the eyes. Also, prolonged sitting leads to neck and back problems.